

The Dominant Pole Approximation

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Introduction and Revision

We give a brief overview of the following topics:

1. The standard second-order transfer function
2. Underdamped systems and examples
3. Higher-order systems and examples
4. The dominant pole approximation (concept)

Demonstration and Analysis

Transfer Functions

We consider a 5th order underdamped system characterised by the transfer function:

$$\begin{aligned} T(s) &= \frac{12614(s+5)(s+6-j8)(s+6+j8)}{5(s+7)(s+5-j9)(s+5+j9)(s+1-j4)(s+1+j4)} \\ &= \frac{12614s^3 + 214438s^2 + 2018240s + 6307000}{5s^5 + 95s^4 + 1135s^3 + 6915s^2 + 22380s + 63070} \end{aligned}$$

Using partial fraction expansion:

$$T(s) = \mathcal{L}^{-1} \left\{ \frac{10.6}{s+7} + \frac{3.747s - 38.1656}{s^2 + 10s + 106} - \frac{114.3472s + 78.8565}{s^2 + 2s + 17} + \frac{100}{s} \right\} \quad (1)$$

After applying the inverse Laplace transform, the expression for the transient response is given by:

$$[10.6e^{-7t} + e^{-5t}(3.747 \cos 9t - 6.322 \sin 9t) - e^{-t}(114.3472 \cos 4t - 8.873 \sin 4t) + 100] u(t)$$

Observing that all poles have real parts at least 5 times greater than the pair of dominant poles closest to the $j\omega$ -axis, we can apply the simplification procedure.

1. Include the pair of dominant poles closest to the imaginary axis (i.e. the smallest real part) in the denominator polynomial. Remove the other poles.
2. Match the DC gain of the original transfer function. That is, multiply the numerator by an appropriate constant so that $T(0) = T_{dp}(0)$.

By applying the dominant pole approximation, we obtain a simplified 2nd order transfer function:

$$\begin{aligned} T_{dp}(s) &= \frac{1700}{(s+1-j4)(s+1+j4)} \\ &= \frac{1700}{s^2 + 2s + 17} \end{aligned}$$

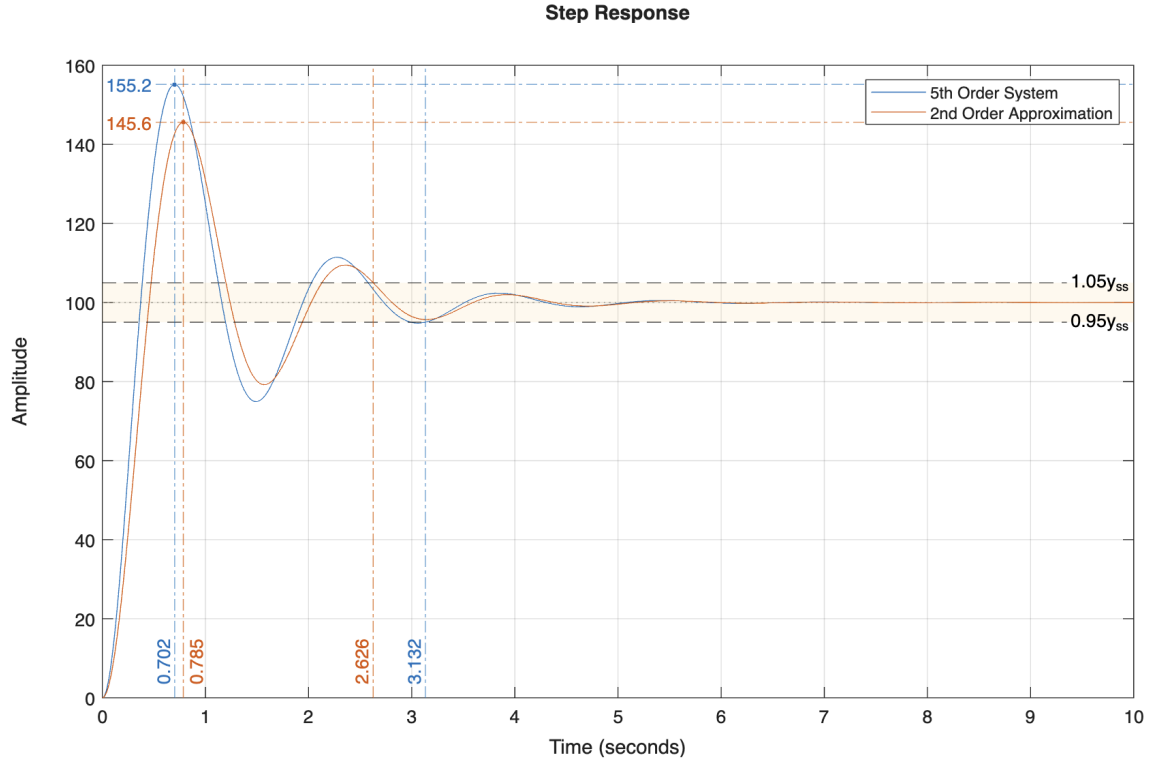


Figure 1: Annotated step response of the original and simplified transfer functions.

Time Domain Analysis

S-Domain Analysis

Comparative Analysis and Discussion

TABLE 3. S-DOMAIN PARAMETERS

Table 1: S-DOMAIN PARAMETERS.

Pole	Natural frequency ω_n	Damped frequency ω_d	Damping ratio ζ
$-1 \pm j4$	$2\pi \cdot 0.656$	$2\pi \cdot 0.637$	0.243
$-5 \pm j9$	$2\pi \cdot 1.639$	$2\pi \cdot 1.432$	0.486
$-7 + j0$	$2\pi \cdot 1.114$	$0 (DC)$	1.000

References

- [1] Dorf, R.C. and Bishop, R.H. (2005) *Modern Control Systems* (13th ed). Section 5.4 "Effects of a Third Pole and a Zero on the Second-Order System Response"
- [2] Ogata, K. (2010) *Modern Control Engineering* (5th ed). Section 5.4 "Higher-Order Systems"

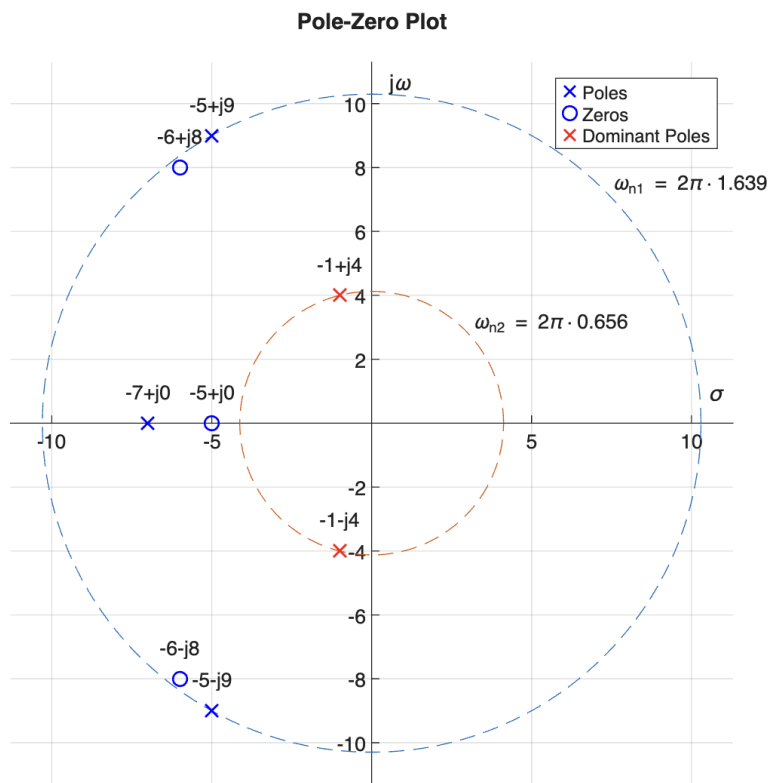


Figure 2: Poles and zeros of the transfer function.